Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.08	20.20	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.02	+0.6%
power (W), mean	627	641	+2.1%
power (W), peak	651	682	+4.8%
energy (Wh)	210	216	+2.7%
energy, parked workers still on at idle power (Wh)	210	216	+2.7%
CPU used (cores), mean	0.857	1.21	+41.8%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.036	0.051	+41.8%
energy per core-hour (Wh)	732	528	-28.0%
node-hours per core-hour	7.00	4.94	-29.5%
pending pods, pod-minutes	0.800	0.000	-100.0%
pending pods, peak	3.00	0.000	-100.0%
HPA replicas, mean	8.77	9.46	+7.9%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	1.05	0.800	-23.8%
requests timed	1,095	1,100	+0.5%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	97.13	94.49	-2.7%
response time (ms), median	85.80	82.30	-4.1%
response time (ms), 95th percentile	211	152	-27.8%
response time (ms), 99th percentile	319	192	-39.9%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.6975	5.6893	-26.1%	[-3.8335, -0.2017]	GAIN
kwh_per_core_hour	0.8011	0.6018	-24.9%	[-0.3746, -0.0241]	GAIN
utilisation	0.0352	0.0497	+41.0%	[+0.0020, +0.0281]	GAIN
pending_pod_minutes_per_hour	2.3040	0.0000	-100.0%	[-6.9120, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	3.0237	2.2857	-24.4%	[-5.2372, +3.8095]	EVEN, no significant difference

What Omni-Compass did

• decisions: 20 (one per minute)

• HPA target writes: 0

• node-pool resizes: 3

• scheduling-floor adds (pending pods): 0

• power-cap pod resizes (in place): 0

• rollout actions: 0

• power cap decided per minute: 0.695

• heat (thermal) per minute: 0.358 0.392 0.421 0.446 0.468 0.488 0.505 0.521 0.534 0.547 0.557 0.562 0.567 0.571 0.577 0.581 0.585 0.588 0.59 0.591

• nodes decided per minute: 6 5 5 4 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3

• HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

```
restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
```

Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	615	618	4.00	5.00	0.000	0.000	0.510	0.599

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	617	624	4.75	7.00	0.000	0.000	0.566	0.763
2	6.00	6.00	616	626	5.00	8.50	0.000	0.000	0.552	0.806
3	6.00	6.00	623	634	5.75	9.00	0.750	0.000	0.712	1.02
4	6.00	6.00	633	644	9.00	10.00	0.000	0.000	0.984	1.26
5	6.00	6.00	632	652	9.00	10.00	0.000	0.000	0.954	1.52
6	6.00	6.00	632	650	9.00	10.00	0.000	0.000	0.939	1.48
7	6.00	6.00	642	672	9.50	10.00	0.000	0.000	1.23	2.06
8	6.00	6.00	644	672	10.00	10.00	0.000	0.000	1.30	2.02
9	6.00	6.00	649	678	10.00	10.00	0.000	0.000	1.39	2.22
10	6.00	6.00	639	656	10.00	10.00	0.000	0.000	1.18	1.89
11	6.00	6.00	614	622	10.00	10.00	0.000	0.000	0.563	0.681
12	6.00	6.00	616	623	10.00	10.00	0.000	0.000	0.539	0.728
13	6.00	6.00	621	632	10.00	10.00	0.000	0.000	0.686	0.956
14	6.00	6.00	632	649	10.00	10.00	0.000	0.000	0.980	1.42
15	6.00	6.00	632	646	10.00	10.00	0.000	0.000	0.985	1.32
16	6.00	6.00	631	645	10.00	10.00	0.000	0.000	0.949	1.30
17	6.00	6.00	624	634	10.00	10.00	0.000	0.000	0.943	1.04
18	6.00	6.00	616	624	10.00	10.00	0.000	0.000	0.552	0.762
19	6.00	6.00	617	622	10.00	10.00	0.000	0.000	0.585	0.710

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.